

MODEL LAB

Naïve Bayes

Aim

To implement and execute the **Naïve Bayes classification algorithm** in Python and predict the class of a new sample.

Algorithm

1. Import the required Python libraries.
2. Create a small training dataset containing features and class labels.
3. Separate the input features (X) and target class (y).
4. Create a **Gaussian Naïve Bayes** classifier.
5. Train the classifier using the training data.
6. Give a new sample as input.
7. Predict its class using the trained model.
8. Display the predicted class.

Program

```
from sklearn.naive_bayes import GaussianNB
```

```
# Training data
```

```
# Features: [Age, Salary]
```

```
X = [  
    [22, 25000],  
    [25, 30000],  
    [28, 35000],  
    [35, 50000],  
    [40, 60000],  
    [45, 70000]
```

```
]
```

```
# Class labels
# 0 = No, 1 = Yes
y = [0, 0, 0, 1, 1, 1]

# Create Naive Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X, y)

# Sample input
age = 32
salary = 45000

# Predict class
prediction = model.predict([[age, salary]])

print("Age:", age)
print("Salary:", salary)

if prediction[0] == 1:
    print("Prediction: Yes")
else:
    print("Prediction: No")
```

Sample Input

Age = 32

Salary = 45000

Sample Output

Age: 32

Salary: 45000

Prediction: Yes

```
... Age: 32  
    Salary: 45000  
    Prediction: Yes
```

Result

The **Naïve Bayes classification algorithm** was successfully implemented in Python. For the given sample input of **age 32 and salary 45000**, the model predicts the class as **Yes**.